TPS-Vis: Interactive Visual Exploration of PD-L1 Expression in Lung Cancer Pathology Images

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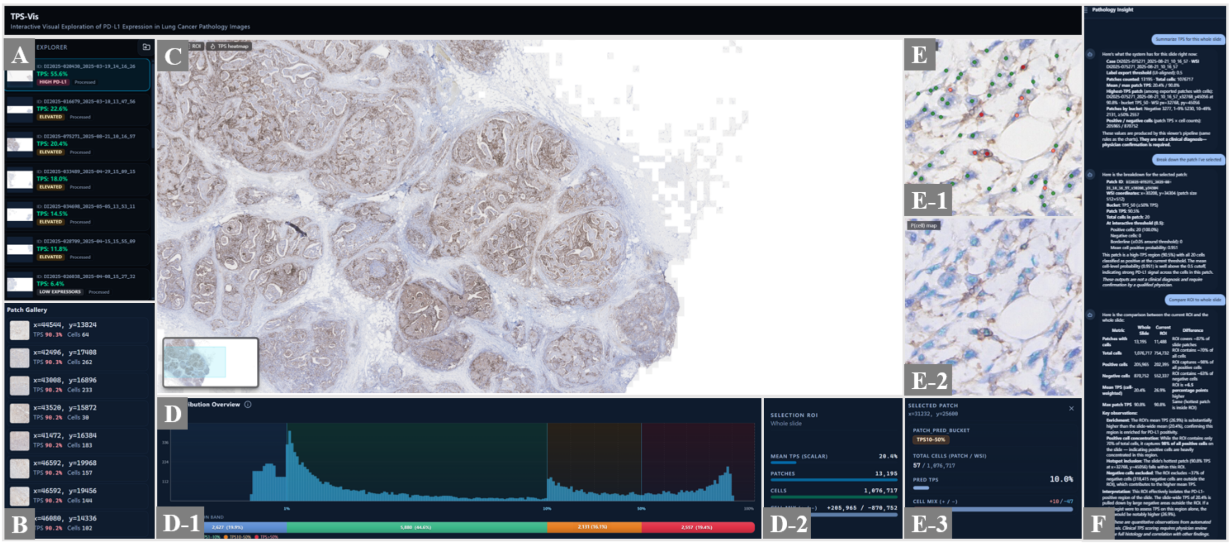

Fig. 1: The interface of our interactive PD-L1 visual analytics system for lung cancer pathology images comprises (A) a specimen explorer, (B) a patch gallery, (C) a multi-scale focus view of the whole-slide image with region of interest (ROI) selection and overlayed TPS heatmap, (D) a cohort-level quantitative dashboard including (D-1) a fine-grained TPS histogram and (D-2) selection-linked summary statistics, (E) a cell-level evidence inspection panel featuring (E-1) cell type maps, (E-2) cell probability maps, and (E-3) detailed patch-level statistics with weighted TPS and cell counts, and (F) the Pathology Insight Agent.

Abstract—In clinical immunotherapy for lung cancer, pathologists commonly assess PD-L1 expression status from IHC slides and use the TPS as an important reference for treatment selection. However, in routine clinical workflows, TPS assessment remains highly dependent on pathologists' experience. Experts need to repeatedly zoom in and out on whole-slide images, locate tumor regions, and inspect membranous staining patterns at the cellular level. This process is easily affected by mixed tumor and non-tumor regions, necrotic tissues, weak positive expression, and other complex pathological factors. To address these challenges, this paper presents an interactive visual analytics system for PD-L1 assessment in lung cancer pathology. The system supports multi-level identification of PD-L1 expression states, presents global and local distributions of positive and negative regions through coordinated multi-scale visualization, and automatically computes TPS. The system helps pathologists perform tumor-region verification, cell-level evidence review, and quantitative TPS assessment more efficiently, thereby improving the consistency, interpretability, and usability of PD-L1-assisted pathological interpretation.

Index Terms—PD-L1 Assessment, Tumor Proportion Score, Digital Pathology, Visual Analytics, Multi-scale Visualization.

Lung cancer is one of the most prevalent and deadly malignancies worldwide, and immunotherapy has become an important component of its clinical treatment. As a critical reference for immunotherapy decision-making, the expression level of programmed death-ligand 1 (PD-L1) is commonly assessed using immunohistochemistry (IHC) slides. Among the quantitative criteria used for PD-L1 interpretation in lung cancer, the tumor proportion score (TPS) is widely adopted. It estimates the proportion of viable tumor cells showing membranous PD-L1 staining among all viable tumor cells [9]. Accurate TPS assessment provides important evidence for patient stratification and treatment selection. Therefore, efficiently and robustly identifying PD-L1 expression patterns in high-resolution pathology slides and assisting physicians in reliable TPS interpretation remain important challenges in digital pathology and clinical immunotherapy.

In clinical practice, TPS interpretation still heavily relies on the experience of pathologists. Physicians usually need to zoom and pan across ultra-large whole-slide images, first locating tumor regions and then examining membranous staining patterns of tumor cells in local areas, before visually estimating the TPS based on their expertise. Unlike a straightforward calculation of the proportion of positive cells, TPS interpretation depends not only on the identification of PD-L1-positive cells, but also on the accurate determination of valid tumor regions and viable tumor cells. Different regions of a slide may contain

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necrotic tissue, inflammatory cell infiltration, and mixtures of tumor and non-tumor components. In addition, weak positive expression, densely distributed cells, variations in staining intensity, and differences among lung cancer histological subtypes further increase the difficulty of interpretation. Such visual and semi-quantitative estimation may lead to inter-observer variability.

In recent years, artificial intelligence methods have advanced rapidly in digital pathology image analysis, demonstrating strong capabilities in tissue segmentation, cell detection, pathological classification, and prognosis prediction [3, 29]. For PD-L1 IHC images, existing methods can perform patch-level expression classification, tumor region recognition, or cell-level positive/negative discrimination [14, 18], and several TPS calculation tools have also been used in digital pathology analysis [10]. However, most existing methods focus on automatic recognition at a single level or on producing a final score, and they provide limited support for multi-level visual tracing and fine-grained TPS interpretation. Pathologists need not only the TPS value, but also an understanding of which tumor regions and local cellular distribution patterns contribute to the score.

Specifically, PD-L1-assisted interpretation still faces several challenges. First, annotation of pathology slides is costly, and cell-level labeling and explicit segmentation are easily affected by image quality, cell overlap, and staining variations. It remains challenging to achieve efficient and robust multi-level state recognition and TPS calculation on whole-slide images. Second, PD-L1 IHC whole-slide images are typically characterized by ultra-high resolution and prominent multi-scale structures, making it important to support coordinated analysis across global distributions, regional structures, and cellular details. Third, a single TPS value cannot sufficiently reveal the spatial evidence and cellular basis underlying the score. Helping physicians trace the source of the score and verify key regions is therefore essential for improving the reliability of assisted interpretation. Accordingly, clinical research urgently requires an interactive PD-L1 expression assessment approach that supports rapid computation and global evidence tracing.

Visual analytics, which integrates data mining, human-computer interaction, and visualization techniques, provides an effective solution to this problem. By combining artificial intelligence models, quantitative statistical analysis, and interactive visualization, visual analytics can transform complex whole-slide images and algorithmic outputs into visual evidence that is understandable, verifiable, and actionable for physicians. Based on this idea, we propose an interactive visual analytics system for PD-L1 expression assessment in lung cancer pathology images. The system supports PD-L1 expression state recognition from the whole-slide level to the cell level, enabling consistent TPS computation. Through coordinated multi-scale views, the system presents PD-L1 expression distributions at the whole-slide, patch, and cell levels to support reliable assessment. Physicians can also manually select regions of interest, after which the system automatically generates TPS-related statistics and visual charts for flexible filtering and on-demand inspection. In addition, the system provides an agent-based interactive question-answering module to improve interpretation efficiency, enabling intelligent, efficient, stable, and interpretable PD-L1-assisted assessment. The main contributions of this work are as follows:

- A multi-level joint modeling method for PD-L1 expression analysis, which simultaneously supports patch-level expression classification and cell-level positive/negative discrimination, enabling efficient multi-scale TPS calculation and cell state recognition.

- An interactive visual analytics system for PD-L1 expression assessment in lung cancer pathology images. The system supports multi-level interpretation from whole-slide images to cellular details, region-of-interest selection, quantitative analysis, and intelligent assessment, enabling efficient interactive exploration of complex pathology slides.

- Experimental validation on real-world lung cancer PD-L1 IHC data. The results demonstrate that the system can intuitively assist experts in slide browsing, screening, and interpretation, improving the consistency and efficiency of the assessment process.

1 Related Work

1.1 PD-L1 Research Progress

In the targeted therapy and prognostic assessment of non-small cell lung cancer (NSCLC), the TPS of PD-L1 has emerged as one of the most critical biomarkers [9]. However, the traditional semi-quantitative assessment performed by pathologists is associated with heavy workloads and substantial inter-observer variability [11]. These difficulties stem from the ultra-large scale of whole-slide images (WSIs), the heterogeneity of the tumor microenvironment, fluctuations in staining intensity, and interference caused by non-specific antibody binding that is common in IHC staining [21]. Furthermore, clinical guidelines from the International Association for the Study of Lung Cancer (IASLC) point out that positive expression in necrotic tissue and in immune cells such as alveolar macrophages can easily interfere with the interpretation of viable tumor cells [23].

To address high-cost cell annotation and the subjectivity of manual assessment, researchers have recently introduced deep learning techniques for end-to-end feature extraction and state recognition in pathology images. Wu et al. proposed a computer-aided diagnosis system based on the U-Net architecture, significantly improving diagnostic reproducibility for novice experts [26]. To eliminate false-positive interference from immune cells, Cheng et al. developed a two-stage workflow deeply coupling a classification network with a YOLO object detection model [4]. Pan et al. further proposed a multi-stage ensemble algorithm combining region-level segmentation (R-Net) with cell-level localization (C-Net), optimizing the synergistic extraction of macro-regional and micro-cellular features [20]. Moreover, to reduce the extreme cost of pixel-level annotation, Lu et al. introduced the clustering-constrained attention multi-instance learning (CLAM) framework based on clustering-constrained attention multi-instance learning. This data-efficient weakly supervised method enables high-precision classification and automated localization of high-value pathological regions using only slide-level labels [15]. Recent clinical validations by Molero et al. and Maniewski et al. further demonstrate that AI-assisted scoring exhibits high sensitivity and specificity at absolute thresholds while substantially reducing reading time [16, 19]. Additionally, a recent review highlights the immense potential of AI algorithms in refining PD-L1 assessment and assisting in the prediction of immunotherapy response [28].

Despite the significant progress in recognition accuracy made by automated algorithms, existing diagnostic systems often exhibit "black-box" characteristics, lacking visual pathological evidence and multi-level contextual explanations. This makes it difficult for physicians to establish sufficient clinical confidence. Addressing this crisis of clinical trust, Giachelle et al. proposed ExaSURE, a visual analytics and explainable AI ecosystem including tools like SKET X. By intuitively presenting and explaining automatically extracted weak annotations and their underlying decision logic through a visual interface, this system effectively enhances algorithmic transparency and physician acceptance [7].

1.2 Pathology Image Visualization

WSIs typically possess ultra-high resolution and multi-scale, multi-level data characteristics, imposing significant computational and cognitive burdens on image processing and analysis. By integrating automated data mining algorithms with interactive interfaces, visual analytics provides an effective path for pathology experts to efficiently understand complex tissue structures [27]. Regarding the design methodology of medical visual analytics systems, Fan et al. recently proposed a design study process model specifically for medical visualization. This model emphasizes the hypothesis-driven decomposition of tasks and offers a comprehensive theoretical framework for the design of clinically customized systems [6].

In terms of multi-scale data rendering and spatial navigation, inspired by the latency bottlenecks of large-scale biological image rendering on the web, Manz et al. developed the Viv multi-scale visualization framework. By leveraging client-side GPU rendering, it achieves pyramidal hierarchical loading of ultra-high-resolution multi-channel data,

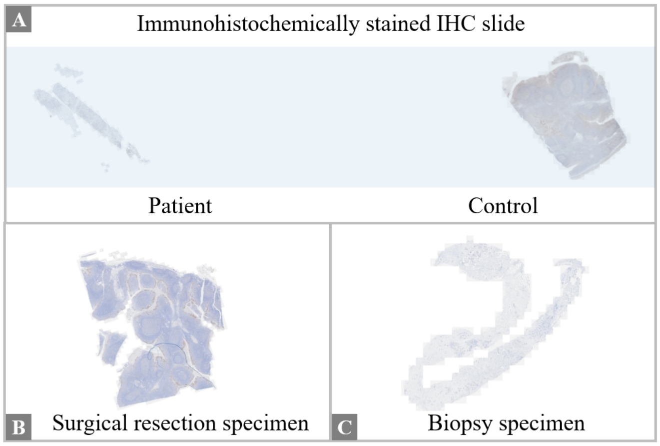

Fig. 2: Example of source images: (A) Immunohistochemically stained IHC slide, (B) Surgical resection specimen, and (C) Biopsy specimen.

ensuring smooth navigation across different scales [17]. Addressing the pathological reading habit of "macro overview followed by micro inspection," the NaviPath system by Gu et al. introduces multi-level AI recommendations and navigational cues, significantly reducing the visual burden on doctors when searching for tiny lesions within complex slides [8]. Furthermore, in 3D and immersive exploration, Veerla et al.'s PathVis system breaks the limitations of traditional displays, utilizing mixed reality technology to achieve large-scale spatial panoramic visualization of WSIs alongside real-time side-by-side AI result comparisons [24].

Regarding the visual mapping of microscopic pathological features and the discovery of complex patterns, the Scope2Screen system by Jessup et al. designed interactive lenses that allow experts to perform multi-channel in-depth inspections and comparisons of local regions of interest while maintaining global structural awareness [12]. Al-Thelaya et al. proposed the HistoContours framework, which ingeniously employs Kernel Density Estimation (KDE) to map discrete microscopic cell classification results into macroscopic contour heatmaps. This effectively alleviates visual clutter in dense cellular regions and supports seamless "contextual shuttling" between macro and micro views [2]. Simultaneously, Al-Thelaya et al. developed the InShaDe framework, which utilizes discrete differential geometry to calculate invariant shape descriptors of cells, providing high-precision quantitative support for morphological clustering [1]. In the exploration of cellular microenvironments, the ImaCytE system designed by Somarakis et al. is specifically used for the interactive analysis of single-cell phenotypes and spatial co-localization patterns [22]; Krueger et al.'s Facetto system combines machine learning with a visual phenotype tree to support hierarchical clustering of multi-channel images [13]; the Visinity system by Warchol et al. utilizes spatial neighborhood analysis to help experts discover and validate cell interaction patterns across slides [25]; and IIComPath by Corvò et al. also establishes a hypothesis-driven visual exploration framework in computational pathology [5].

In this field, the recent work of Xu et al. (2025) is highly representative and serves as a focal point. They constructed an intelligent interactive visual analytics system for ultra-large-scale pathological images. The system innovatively introduces diffusion models to achieve high-precision multi-modal tissue segmentation and designs a multi-scale visualization mechanism that supports multi-window synchronous linkage and dynamic filtering of tissue components. This paradigm, which deeply synergizes complex AI-extracted features with efficient expert interaction, greatly enhances the interpretability of pathological features and the efficiency of group-level comparisons [27].

2 Research Background and Motivation

This study was motivated by our collaboration and discussions with several experts in cancer pathology. Our collaborators include a clinical pathologist who have long been engaged in the pathological diagnosis and prognostic assessment of various cancers, as well as a pathology image analysis expert whose research focuses on digital pathology image analysis, whole-slide image modeling, tissue- and cell-level recognition, and intelligent analysis methods. In clinical pathology practice, experts usually assess PD-L1 expression on immunohistochemically stained slides according to the staining appearance, thereby evaluating the pathological status of the tissue. Fig. 2A shows a schematic example of a clinically used IHC slide. The length and width follow the actual configuration of a clinical glass slide, with multiple tissue fragments distributed on the left and right sides and a large blank area in the middle. One side of the slide contains the patient specimen to be assessed, while the other side contains control tissue. The purpose of the control tissue is not to conduct a direct differential comparison with the patient specimen, but to determine whether the IHC staining is adequate, whether the antibody localization is accurate, and whether positive expression can be reliably detected, thereby excluding false-negative results caused by insufficient staining sensitivity. In general, the control tissue is not necessarily normal tissue; it may also be known tumor-positive tissue.

After the staining quality is confirmed, pathologists visually estimate the proportion of stained regions in the patient tissue and describe the result using coarse percentage intervals, such as 5%, 10%, 20%, and 50%, to characterize the severity of tumor-related expression and support treatment decision-making. However, patient specimens can present diverse morphologies due to different sampling procedures. For example, surgical resection specimens, as shown in Fig. 2B, usually appear as tissue blocks, whereas needle biopsy specimens, as shown in Fig. 2C, often appear as elongated strips. During embedding and sectioning, biopsy tissues may be divided into multiple fragments and distributed across different positions on the slide. The assessment of IHC slides requires pathologists to first visually locate tumor regions and then verify them through zooming and close inspection. This process is strongly affected by the morphology of pathological tissue. In particular, needle biopsy specimens are often curved, irregular, and fragmented after embedding and sectioning, making it difficult to estimate the proportion of PD-L1-positive expression. As a result, considerable inter-observer variability may occur among different pathologists. Therefore, TPS has been introduced to more precisely quantify the positive proportion of the specimen. Nevertheless, several challenges remain. On the one hand, the relative positions of the patient specimen and the control tissue on the slide are not fixed, and their morphologies vary substantially, requiring experts to make careful judgments based on each individual slide. On the other hand, although existing tools can perform TPS assessment on whole-slide IHC images, a single TPS value is insufficient to fully characterize the pathological status of the tissue. In addition, necrotic tissue, weak positive expression, and the mixture of tumor and non-tumor cells in IHC slides can all affect the stability and reliability of the final score.

IHC slides are characterized by dispersed tissue distributions, large blank regions, the coexistence of patient specimens and control tissues, and substantial morphological differences between biopsy and surgical specimens. At present, there remains a lack of interactive visual analytics systems that support multi-level evidence tracing and detailed assessment across whole-slide images, which is essential for improving the efficiency, stability, and reliability of clinical PD-L1 expression interpretation.

3 Design Goals

We focus on the clinical analysis of PD-L1 expression in lung cancer and define the following four key design goals:

DG1: Multi-level Recognition. PD-L1 prognostic assessment requires not only an overall judgment of the distribution of positive and negative expression regions on a slide, but also fine-grained identification of membranous staining status at the cellular level, so as to support more reliable positive/negative determination. However, physicians cannot provide detailed annotations for ultra-large images, and the high cost of cell-level annotation and explicit segmentation workflows also limits their application in large-scale inference. Therefore, rapid state recognition at both the region and cell levels is required to achieve accurate identification and high-confidence assessment of PD-L1 expression status.

DG2: Multi-scale Visualization. PD-L1 IHC slides exhibit prominent multi-scale characteristics. Whole-slide and region-level images enable the observation of macroscopic staining and expression distributions, whereas cell-level images help identify cell membrane staining, local heterogeneous regions, and cell boundary details. In clinical slide reading, physicians need to first locate tumor regions before analyzing relevant indicators. Switching between images at different scales helps physicians perform global screening and local confirmation. Therefore, the system needs to provide a multi-scale visualization mechanism with synchronized coordination between region-level and cell-level views to support efficient observation.

DG3: Automated Quantitative Analysis. Physicians usually need to focus on regions of interest, and selecting and automatically analyzing the immune expression within these regions can facilitate rapid and refined assessment of tumor status. However, due to tissue heterogeneity, the mixture of tumor and non-tumor cells, variations in staining intensity, and the difficulty of distinguishing weakly positive borderline cases, TPS calculation and assessment face challenges such as noise interference, cross-scale modeling difficulty, and insufficient interpretability. Therefore, the system needs to automatically compute and statistically analyze tissue components and related expression indicators, thereby improving the objectivity of the assessment process and the interpretability of the results.

DG4: Efficient Interactive Exploration. PD-L1 whole-slide images are typically large in scale and information-dense. Experts need to continuously locate regions of interest, filter key information, and compare analysis results across different levels within complex tissue regions. This process is time-consuming and prone to visual burden, especially in weakly stained regions, densely cellular regions, and highly heterogeneous regions. Therefore, the system needs to provide efficient interactive exploration capabilities, supporting rapid navigation, flexible filtering, coordinated result exploration, and on-demand detailed inspection, thereby improving the overall efficiency of the assessment.

4 Data Description

The dataset used in this study comprises 20 PD-L1 IHC whole-slide images of lung cancer. After scanning, the level-0 images have an average size of approximately $73{,}340 \times 37{,}700$ pixels. By tiling at level 0 with a patch size of 512×512 pixels and filtering out empty background regions, we obtained a total of 63,194 IHC patches. The dataset covers not only large surgical resection specimens but also the clinically more challenging needle biopsy specimens, which are typically thin, elongated, and prone to fragmentation. It therefore spans diverse tissue morphologies, heterogeneous spatial distributions, and varying levels of PD-L1 expression, enabling a thorough evaluation of the robustness and practical utility of the system. With a total pixel count reaching the hundred-million scale, the dataset exemplifies the typical ultra-high resolution and multi-scale characteristics of digital pathology data.

5 Methods

5.1 Overall Workflow

As shown in Fig. 3, the overall workflow of the system comprises multi-level expression modeling, automated quantitative analysis, and interactive visual interpretation. In Phase I, high-resolution whole-slide images are decomposed into pixel patches, and a cell-aware weakly supervised distillation framework performs patch-level TPS grading and cell-level state recognition in parallel. In Phase II, the system uses spatial indexing to automatically filter invalid blank backgrounds and, based on clinical thresholds, performs automatic bin-wise computation and statistical aggregation of TPS indicators over patches within valid tumor regions. In Phase III, the system provides a multi-scale coordinated interactive interface that, through region of interest (ROI) selection and cell-level detail cards, along with an agent to assist pathologists in lesion localization and evidence tracing, thereby enhancing the consistency and interpretability of PD-L1 assessment.

5.2 PD-L1 Expression Status Classification

The algorithm underlying our system is a cell-aware weakly supervised modeling method for automatic TPS assessment on PD-L1 IHC images. Its core objective is to extract TPS-related cell-level evidence from pathology images and to further perform patch-level TPS grading, without requiring manual per-cell annotation during training or full instance segmentation at inference time.

In the training stage, the algorithm first leverages cell instance masks together with a pathology pre-trained model to construct teacher-side cell representations. Specifically, let F^T denote the feature map extracted by the teacher network, and let $\tilde{M}_i$ denote the mask region of the i-th cell instance in the feature-map coordinate system. The teacher representation of this cell is then obtained by average pooling of the features within the mask region:

$$z_i^T = \frac{1}{|\tilde{M}_i|} \sum_{(u,v)\in\tilde{M}_i} F^T(u,v),$$

where z_i^T denotes the teacher-side feature vector of the i-th cell, (u,v) denotes a spatial coordinate on the feature map, $F^T(u,v)$ denotes the teacher feature vector at that location, and $|\tilde{M}_i|$ denotes the number of feature points inside the cell mask. Subsequently, the algorithm distills this multi-stage cell feature extraction process into a lightweight student network, and uses a point-supervised localization head to predict cell centers, based on which cell-level features are sampled at the predicted cell positions.

In the downstream prediction stage, the cell features within a patch are organized as a variable-length sequence and fed into a dual-task Transformer with a shared encoder. The model has two output branches: the cell-level branch predicts the PD-L1 negative/positive status of candidate tumor cells, while the patch-level branch predicts the corresponding TPS grade of the patch. Unlike methods that directly apply a threshold over the proportion of positive cells, the proposed algorithm uses the Transformer to model inter-cell relations, spatial context, and the overall staining distribution, and employs a cross-scale consistency constraint to align the cell-level positive ratio with the patch-level TPS prediction. The overall training objective is:

$$\mathscr{L} = \mathscr{L}_{cell} + \mathscr{L}_{patch} + \lambda_{cons}\mathscr{L}_{cons}, \quad \mathscr{L}_{cons} = \left| \hat{t} - \frac{1}{N}\sum_{i=1}^{N} p_i \right|,$$

where $\mathscr{L}_{cell}$ is the cell-level PD-L1 negative/positive discrimination loss, $\mathscr{L}_{patch}$ is the patch-level TPS supervision loss, $\mathscr{L}_{cons}$ is the cross-scale consistency loss, and λ_{cons} is the corresponding weight. $\hat{t}$ denotes the TPS estimate predicted by the patch branch, p_i denotes the positive probability of the i-th cell predicted by the cell branch, and N denotes the number of cell candidates within the patch. The consistency loss uses the L_1 distance to enforce agreement between the patch-level TPS estimate and the cell-level positive-ratio estimate. As a result, in a single forward pass the model simultaneously outputs cell locations, cell-level PD-L1 states, and patch-level TPS classification results. Experimental results show that, compared with traditional DAB-threshold methods, hand-crafted features, general-purpose IHC quantification tools, and open-source PD-L1 pipelines, the proposed method achieves more stable performance on the four-class patch-level TPS task, while preserving cell-level evidence that is directly aligned with the pathological interpretation process.

5.3 Visual Design and Interaction

5.3.1 Data Selection View

The core design goal of the left sidebar (Fig. 1A, B) is to provide an efficient entry point into the data and to prevent users from searching blindly through a large case pool. This module is composed of two vertically stacked components.

Specimen Explorer. During the data import stage, the system parses the image-pyramid manifest of each whole-slide image on the server side and generates case cards in the upper-left panel. To improve clinical screening efficiency, cases are by default sorted in descending order

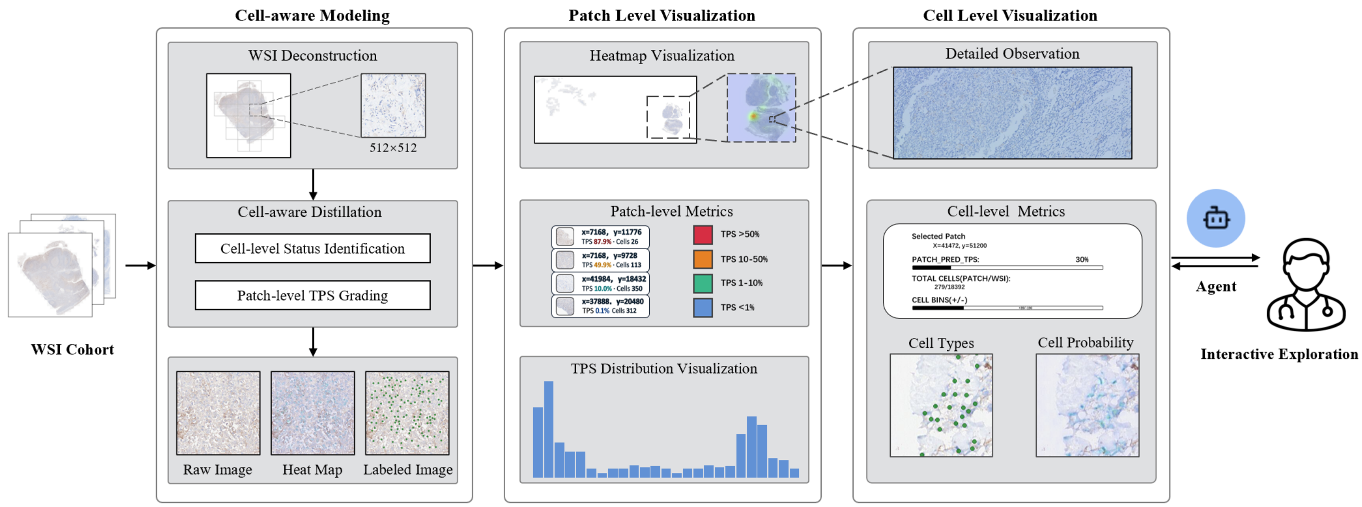

Fig. 3: The visual analytics pipeline of our PD-L1 expression evaluation system is composed of multi-level representation modeling, patch-level quantitative analysis, and cell-level interactive exploration stages.

by their average TPS scores on the client side, and are visually encoded with status phrases (e.g., *High PD-L1*, *Elevated*, *Low expressors*) and high-contrast borders (Fig. 1A), so that high-expression cases stand out at a glance.

Patch Gallery. Located in the lower half of the left column (Fig. 1B), the Patch Gallery serves as a retrieval hub bridging macroscopic and microscopic scales. The system first filters out background patches that contain no cell-level data and retains only valid tumor regions. The gallery is equipped with a built-in bin-wise filter based on clinical scoring standards. When the user clicks on a specific high-risk patch, the system triggers a fly-to interaction that smoothly pans and zooms the main view to the corresponding spatial location. In addition, the Patch Gallery maintains a tight linkage with the ROI selection in the central panel: let the set of valid patches on the whole slide be P and the current selection be ROI; the gallery only renders the intersection $P \cap \mathrm{ROI}$ and dynamically reorders these patches according to their predicted TPS value τ.

5.3.2 IHC Image View and TPS Distribution View

The central panel (Fig. 1C, D) occupies the largest amount of screen space and serves as the core slide-reading and quantitative-analysis hub of the system. It is designed to address two key pain points encountered when navigating ultra-high-resolution images: spatial disorientation and the difficulty of quantifying heterogeneity.

IHC Image View. The upper part of the central panel builds a smooth image-pyramid rendering engine based on OpenSeadragon (Fig. 1C). To provide visual shortcuts without occluding the underlying tissue morphology, the system uses kernel density estimation to overlay a high-contrast TPS heatmap on the WSI surface. In addition, users can frame regions of interest in the tumor microenvironment using a local ROI tool. The underlying algorithm automatically snaps the user-drawn rectangle to the 512×512 pixel patch grid and extracts feature data within the spatial intersection in real time.

TPS Distribution View. The lower part of the central panel provides a highly customized TPS quantitative dashboard (Fig. 1D). Through the TPS distribution histogram and the selection-linked summary statistics, it enables multi-dimensional data characterization. For each patch with predicted TPS score $\tau \in [0,1]$ output by the model, the system maps it to a fine-grained 0.1% histogram bin:

$$k = \mathrm{clamp}(\mathrm{round}(1000 \cdot \tau), 0, 1000).$$

To make the visual distribution of the histogram align with the four clinical TPS thresholds, the system introduces a density-aware scale mapping on the horizontal axis. Let n_b be the number of patches in clinical bin $b \in \{\mathrm{Neg}, \mathrm{T1}, \mathrm{T10}, \mathrm{T50}\}$, and let N_{tot} be the total number of valid patches. The pixel width W_b allocated to each clinical bin within the total plotting width W_{plot} is given by:

$$W_b = \begin{cases} \dfrac{n_b}{N_{\mathrm{tot}}} W_{\mathrm{plot}}, & N_{\mathrm{tot}} > 0, \\ \dfrac{1}{4} W_{\mathrm{plot}}, & N_{\mathrm{tot}} = 0. \end{cases}$$

This mapping ensures that the total mass of the histogram bars within each background band exactly matches the true proportion of patches in the corresponding clinical bin. To prevent local extremes from causing visual saturation along the vertical axis, the system further introduces a dynamic filling ratio ρ (whole-slide mode: $\rho = 0.9$; ROI mode: $\rho = 0.8$) to determine the upper bound of the Y axis:

$$Y_{\mathrm{max}} = \frac{\tilde{c}_{\mathrm{max}}}{\rho}.$$

With this dashboard, users can sensitively detect the spatial heterogeneity of the tumor microenvironment by examining whether the histogram exhibits a bimodal structure.

5.3.3 Cell-Level View

The right sidebar (Fig. 1E) is designed to open up the black box of the algorithm and to provide solid underlying data-interpretation support, thereby helping pathologists build trust in the quantitative scoring.

Multi-layer Preview Strip. When the main view smoothly navigates to a target patch, the preview module in the upper-right corner automatically and synchronously loads the high-resolution image of that region (Fig. 1E). This component integrates the cell tpye map (*Cell labels*, Fig. 1E-1) and the cell probability map (*P(cell) map*, Fig. 1E-2), and supports independent zooming and panning within the mini-viewport. The two layers are linked at the pixel level in real time, allowing experts to perform multi-dimensional detailed verification of local lesions without disturbing the global context of the main viewport, thereby alleviating the cognitive load caused by frequent layer switching.

Cell-Level Detail Card (Selected Patch). Below the preview strip lies an information-rich statistical panel (Fig. 1E-3). This panel asynchronously loads the precise coordinates and classification probabilities of all cells within the current patch in real time. It uses stacked bar charts to intuitively compare the absolute counts of negative and positive tumor cells, and computes the local cell-weighted TPS. More importantly, it also displays the weight contribution of the cell base of the current patch relative to the total cell base of the whole slide.

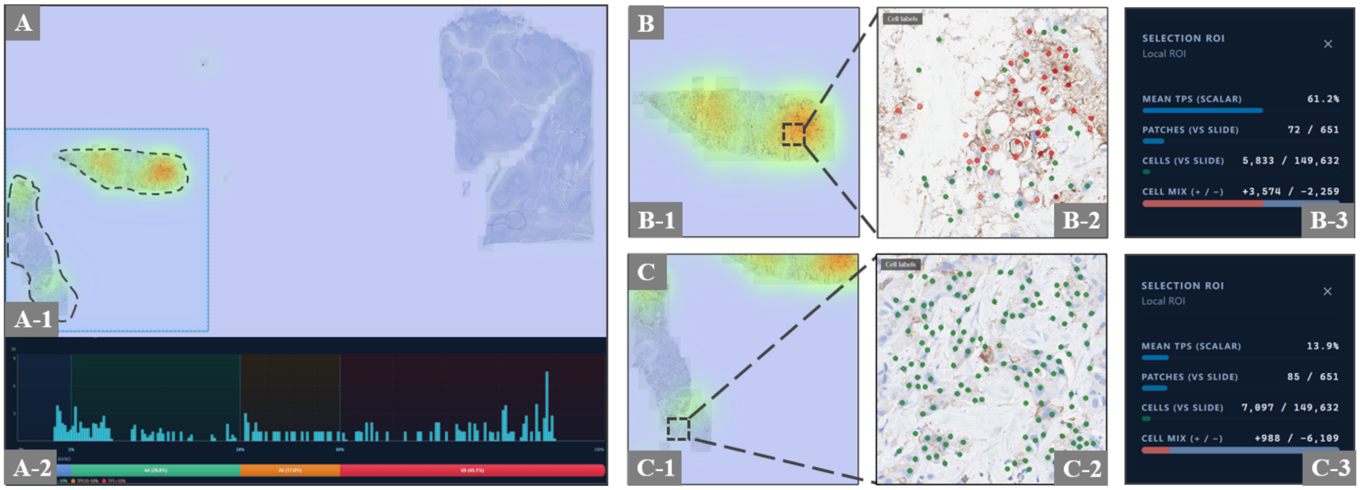

Fig. 4: Assessment of Biopsy Heterogeneity: (A) Global Overview View, (B) High-expression Region (Strip A), (C) Low-expression Region (Strip B)

By juxtaposing "microscopic expression intensity" and "macroscopic component weight", this design forms a complete chain of evidence tracing and effectively corrects the visual bias caused by local high expression.

5.3.4 Pathology Insight Agent

To improve the information-access efficiency and medical interpretability of the system, we introduce the *Pathology Insight Agent* (Fig. 1F) as a unified natural-language question-answering layer, mainly designed to help pathologists understand complex quantitative indicators and spatial heterogeneity features. At the architectural level, we deliberately decouple complex interaction orchestration and free-form report generation, and adopt a lightweight hierarchical strategy of "outer-layer semantic parsing plus inner-layer controlled retrieval."

At the outer layer, a general-purpose large language model is integrated as the natural-language parsing module. The current implementation uses the DeepSeek model to perform semantic understanding and intent recognition on user queries, and to map them precisely into structured context parameters such as the current case, the patch under inspection, the local ROI selection, and the clinical thresholds for cell-level positivity.

At the inner layer, to ensure rigor in medical scenarios and to fundamentally eliminate hallucinations from the large language model, the agent is strictly constrained in terms of system-operation and orchestration permissions. Its core function is limited to fact-based objective question-answering. After completing the request mapping, the system issues queries to the underlying database and back-end computation services through controlled data interfaces, retrieving unique raw statistical snapshots such as the patch with the highest TPS on the entire slide or the comparison between the cell counts within the local ROI and those across the whole slide.

Users can engage in conversational question-answering with the agent. It should be emphasized that, in order to ensure the safety of medical decision-making, we pre-define a set of core questions that pathologists are likely to focus on during clinical reading, together with their pre-computed objective answers. The system's responses are built on top of these unique, objective, pre-computed values: the agent reads structured output fields and uses fixed textual templates to organize the final answer in a fill-in-the-blank manner. The pathology insights produced by the system are thus standardized restatements of the underlying numerical results from the analysis pipeline, rather than free-form scientific diagnostic conclusions generated by the language model or subjective inferences drawn from a vision-language model's interpretation of the image content. This design provides a flexible conversational experience while preserving a clear safety boundary for assisted interpretation.

6 CASE STUDIES

To evaluate the effectiveness of the system in the clinical interpretation of lung cancer PD-L1, we conducted case studies on 20 real IHC whole-slide images, focusing on the system's ability to support multi-level evidence tracing, spatial heterogeneity analysis, and assisted research hypothesis generation. The case evaluation involved two collaborating domain experts: E1 is a chief pathologist with more than ten years of clinical experience, and E2 is an academic researcher specializing in computational pathology. They provided hands-on feedback and evaluations from multiple dimensions, including clinical workflow adaptability, result consistency, interaction efficiency, and system interpretability.

6.1 Case1: Assessment of Biopsy Heterogeneity

PD-L1 expression in the tumor microenvironment often exhibits substantial spatial heterogeneity, and foci of very low and very high expression may coexist within the same biopsy specimen. Relying solely on globally aggregated indicators tends to mask clinically meaningful local high-risk regions. Such heterogeneity is particularly critical in small needle biopsy specimens, where the limited sampling area makes the assessment highly sensitive to "which fragment happens to be sampled," giving rise to a substantial risk of interpretation error.

This case study presents a typical clinical interference scenario: the left side of the slide contains the patient biopsy, while the right side contains the control tissue (Fig. 4A-1). Because the patient's biopsy tissue is too long along the scanning direction, it is cut into two adjacent regions (*Strip A* and *Strip B*) during physical sectioning and grid partitioning. The global statistics show that the overall TPS on the patient side falls within the critical boundary region of 35.7%, while the local expression diverges dramatically: one strip reaches as high as 61.2%, whereas the other is only 13.9%.

At the beginning of the analysis, the TPS distribution view at the bottom of the central panel provides global data cues. This view shows the histogram aggregated at 0.1% bin granularity and rescaled by the density-aware mapping, strictly aligned with the four clinical background bands. The slide-level TPS allocation strip macroscopically presents the proportions of the different clinical bins. In this view (Fig. 4A-2), the histogram exhibits a typical bimodal structure: one main peak is concentrated in the extremely low expression region around 1%, while a secondary peak appears in the high expression range of 50%–100%. This atypical shape constitutes a key visual cue and guides the analysis toward a spatial decomposition based on the heatmap. Fine-grained spatial analysis on the heatmap reveals a clear topological difference: the two tissue fragments on the left side exhibit drastically different staining patterns. The upper region shows a prominent red high-contribution signal, indicating strong PD-L1 aggregation,

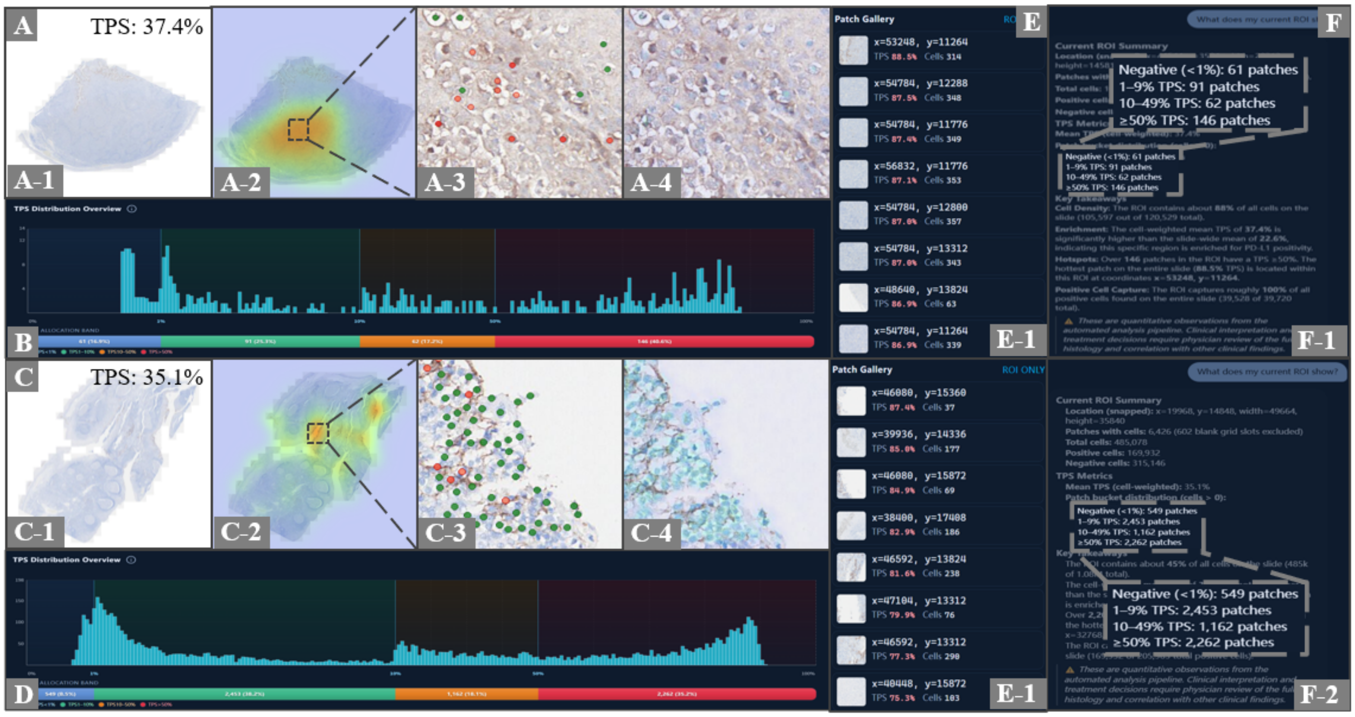

Fig. 5: Comparative Analysis of Boundary Cases:(A, C) pathology insights, (B, D) TPS Distribution Overview,(E)Patch Gallery, (F) Pathology Insight Agent

whereas the lower region is dominated by a large blue area, indicating a predominance of negative regions.

The user then employs the local ROI tool to perform targeted selections on the two biopsy fragments. Through the multi-scale coordinated views, the difference between the two ROIs becomes immediately apparent: when *Strip A* is selected, the Selection ROI panel reports a locally weighted average TPS of 61.2%, and the Patch Gallery is filled with high-expression patches (Fig. 4B); when *Strip B* is selected, the local TPS drops sharply to 13.9% (Fig. 4C). Finally, the user moves the viewport to the control tissue on the right side and browses it within the same interaction paradigm as a low-expression reference context.

This multi-dimensional comparison effectively strengthens the chain of evidence linking the local high-expression peak on the patient side to the large negative cell base. From a research perspective, such differences in expression distribution not only reflect intra-tumoral spatial heterogeneity but also directly reveal their influence on the stability of boundary-case interpretation. Identifying "high-contribution regions" and "controversial regions" through the visual analytics system substantially improves the interpretability and confidence of PD-L1 assessment.

6.2 Case2: Comparative Analysis of Boundary Cases

In an exploratory analysis, we compared two cases located near the clinical decision boundary and observed that, despite sharing similar global statistical features, they exhibit markedly different spatial distribution patterns. This observation motivates a further investigation into the potential association between PD-L1 spatial topology and clinical prognosis. As shown in Fig. 5, Case A and Case C are highly consistent at the macroscopic statistical level. To quickly verify their quantitative indicators, the user invokes the Pathology Insight Agent for natural-language-driven on-demand retrieval. By asking "What does my current ROI show?", the agent accurately parses the user's spatial intent and rapidly retrieves the patch distribution across different TPS bins: Case A contains 146 patches in the $\geq 50\%$ TPS bin, yielding a weighted average TPS of 37.4% within this region (Fig. 5F-1); Case C, on the other hand, contains 2,262 patches in the same bin, corresponding to a weighted average TPS of 35.1% (Fig. 5F-2). Although the patch counts differ by an order of magnitude, the frequency distributions of their TPS histograms across the clinical bins are largely consistent (Fig. 5B and D). The user can further filter high-TPS samples through the Patch Gallery (Fig. 5E) and inspect cell-level algorithmic annotations (Fig. 5A-3 and C-3) for visual verification.

However, when fine-grained spatial analysis is performed using the heatmap provided by the system, a striking topological difference emerges: Case A presents a typical single-center focal distribution, with only one concentrated red high-contribution region (Fig. 5A-2), whereas Case C exhibits a clearly multi-centric and dispersed distribution, with three independent high-expression clusters in the heatmap (Fig. 5C-2).

Based on these observations, we formulate the following hypothesis: under similar total tumor burdens, the spatial aggregation pattern of PD-L1 expression may reflect different evolutionary states of the tumor microenvironment. In particular, a multi-centric and dispersed expression pattern may indicate broader immune infiltration or more complex clonal heterogeneity, and may therefore carry potential biomarker value for predicting subsequent immunotherapy response and patient recovery. The formulation of this hypothesis demonstrates the system's capability to support research that goes beyond traditional numerical assessment and that uncovers higher-order spatial topology features.

7 Expert Feedback

The two collaborating domain experts (E1 and E2) provided feedback on our visualization work. Both experts agreed that coordinated multi-scale visualization combined with the automated quantification pipeline substantially mitigates the subjective bias inherent in conventional "eyeball" estimation on whole-slide immunohistochemistry (IHC) images. E1 emphasized that linking quantitative summaries to regions of interest (ROIs) makes subtle expression fluctuations in needle biopsy material easier to capture and compare across fragments, rather than collapsing them into a single impressionistic score. For cases hovering near clinically critical TPS cutoffs (e.g., 1% or 50%), the multi-level chain of evidence—from patch-level distributions to cell-level maps—was described as particularly valuable: it reduces the risk of misclassification driven by visual fatigue, heterogeneous staining, or ambiguous background signal, and supports a clearer shift from purely experience-based judgment toward data-grounded review. E1 also noted that explicitly tying numeric summaries to spatial context made disagreements easier to articulate and resolve during joint review.

From a research-oriented perspective, E2 highlighted the system's strength in surfacing spatial topology patterns that are difficult to infer from global TPS alone. In particular, juxtaposing cases with similar aggregate TPS but divergent spatial layouts helped articulate new hypotheses about how PD-L1 spatial heterogeneity might relate to immunotherapy response and microenvironmental evolution. E2 summarized this as follows: "The intuitive quantification of heterogeneity goes beyond a single number; it offers fresh visual evidence for reasoning about how the tumor microenvironment may be evolving."

Regarding efficiency, both experts reported that the Pathology Insight Agent and the multi-layer preview strip noticeably reduced the cognitive overhead of the repeated zooming and panning on ultra-large IHC slides. In informal timed sessions on a subset of complex cases, they estimated that end-to-end review time dropped by roughly 40% relative to their usual digital-slide workflow while maintaining high perceived consistency; these figures are indicative only and were not collected under a formal controlled study design. Overall, the feedback supports treating the system as a practical adjunct for exploratory review and hypothesis generation, with final clinical decisions remaining the responsibility of qualified domain experts.

8 Discussion and Limitation

In the context of clinical PD-L1 assessment for lung cancer, large-scale and multi-level pathology image analysis is critical for immunotherapy decision-making. By integrating a cell-aware weakly supervised learning model with interactive visual analytics, this work enables accurate quantification and visual tracing of PD-L1 expression states and TPS indicators. Following a modular design principle, the system deeply couples the weakly supervised distillation framework at the algorithmic level with the multi-scale coordination mechanism at the front-end. Unlike existing AI diagnostic systems that focus on a single numerical output, our system emphasizes *evidence traceability*. Through the Pathology Insight Agent and the Cell-Level View, microscopic features identified by the algorithm are transformed into visual evidence that pathologists can verify. This significantly improves interpretation confidence when handling complex clinical scenarios that are common in PD-L1 expression analysis, such as weak positive interference and the mixture of tumor and immune cells. The system also demonstrates strong robustness across specimens of different morphologies, including needle biopsy and surgical resection samples. The case studies show that the spatial heterogeneity topology features revealed by the system not only help improve the stability of TPS scoring but also assist pathologists in generating new research hypotheses regarding tumor microenvironment evolution. Moreover, the design of the system fully takes into account the practical hardware conditions of clinical environments. Through a modular rendering architecture and a hierarchical preprocessing mechanism, the system delivers a smooth interactive experience even on ordinary mobile workstations. Experimental tests show that the system can run stably on a standard laptop equipped with a mainstream mobile processor (e.g., Intel Core i5-13500H) and 16 GB of memory. This low dependence on hardware significantly lowers the deployment barrier of AI-assisted pathology analysis, allowing seamless integration into primary hospitals or clinical settings with limited resources.

Despite the significant improvements in PD-L1 assessment efficiency and consistency, the system still has the following limitations. (i) The current system mainly focuses on single-modality IHC image analysis. In clinical practice, multi-modal comparison with co-registered H&E-stained images is highly valuable for accurately distinguishing tumor cells from complex stromal backgrounds. Future work will explore multi-modal registration and overlay analysis. (ii) Although the system introduces a natural-language agent, the interaction still relies primarily on pre-defined question–answer templates and controlled retrieval, leaving room for improvement in terms of flexibility and cross-case knowledge-graph reasoning.

9 Conclusion

This paper presents an interactive visual analytics system for PD-L1 expression assessment in lung cancer pathology, aiming to address pain points in the clinical workflow such as low efficiency, strong subjectivity, and the difficulty of quantifying spatial heterogeneity in TPS interpretation. The system innovatively integrates a multi-level expression modeling algorithm with a multi-scale coordinated visualization mechanism, supporting seamless transitions from whole-slide overviews to cell-level details. Through the Pathology Insight Agent and the Cell-Level View, the system not only improves the consistency and efficiency of clinical interpretation but also enhances the interpretability of AI-assisted decision-making via a chain of visual evidence. The case studies demonstrate the system's significant potential in revealing intra-tumoral spatial heterogeneity and in supporting the generation of research hypotheses. Future work will focus on the joint analysis of multi-modal data (e.g., IHC and H&E staining) and on further validating the generalization of the system in multi-center clinical scenarios.

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